

$f: \mathbb{R} \rightarrow \mathbb{R}$, a Mark-a

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

diffable

$$f(x) = x^3 + 5x^2 + 5$$

$f: \mathbb{R} \rightarrow \mathbb{R}$, $x=a$

$$f'(x) = 3x^2 + 10x$$

$\rightarrow f'(a) = 0$ is a critical poi-

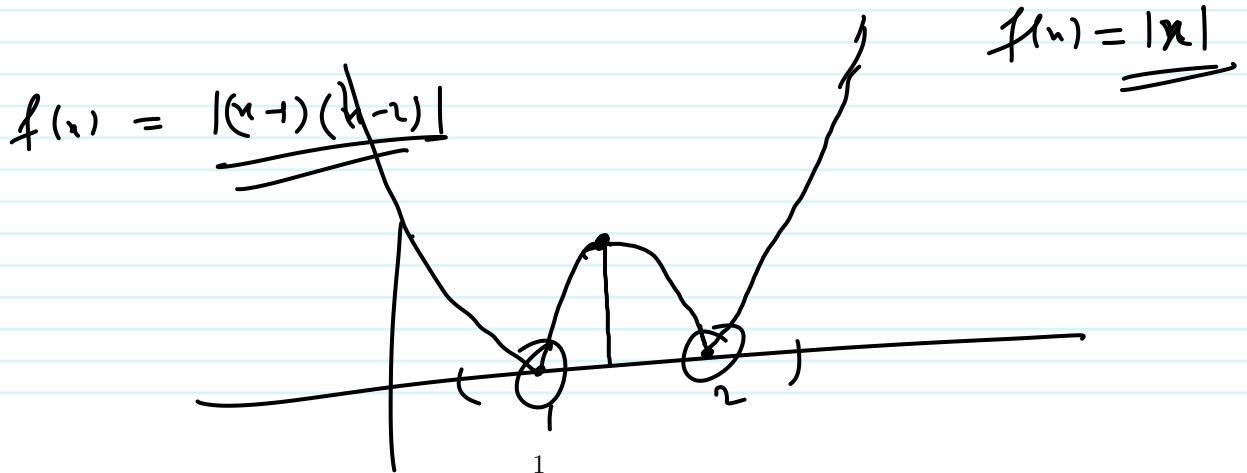
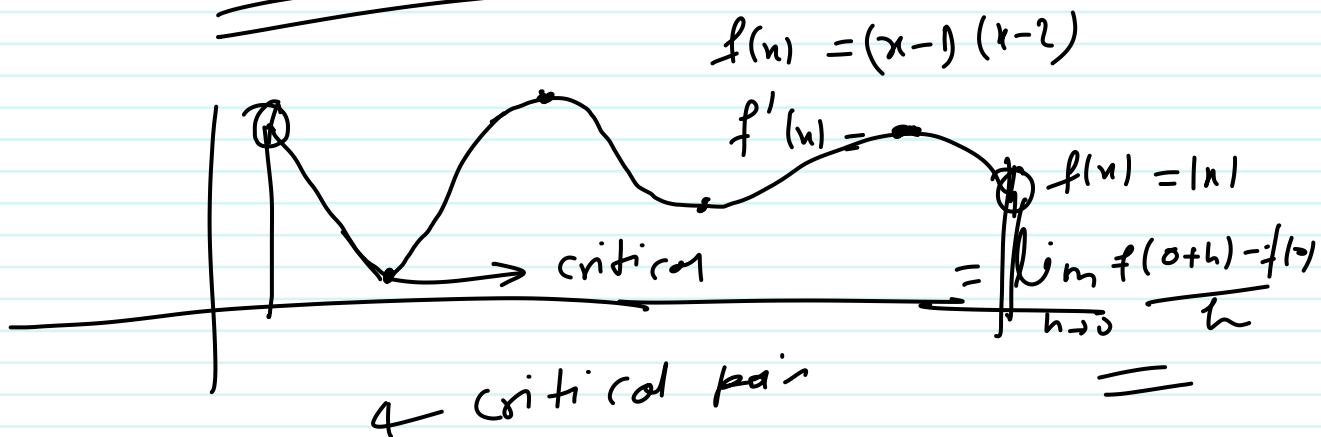
$\rightarrow$ if not diff at a then a is a critical

$$f(x) = 5$$

$$f'(x) = 0$$

$$f(x) = |(x-1)(x-2)|$$

$$f(x) = |x| \text{ at } 0$$



$$f(x) = \frac{(x-1)(x-2)}{2x-3} = \frac{x^2 - 3x + 2}{2x-3} = -b/2a = \frac{3}{2}$$

$|x-1|(x-2)|$

$$f(x) = \begin{cases} (x-1)(x-2) & x < 1 \\ -(x-1)(x-2) & 1 \leq x \leq 2 \\ (x-1)(x-2) & x > 2 \end{cases}$$

$$f'(x) = \begin{cases} 2x-3 & x < 1 \\ -(2x-3) & 1 < x < 2 \\ 2x-3 & x > 2 \end{cases} \quad (-\infty, 1)$$

$f'(x) = 0$ $2x-3=0 \Rightarrow x = 3/2$ X

α, β $f(x) = x^3 + 5x^2 + 2x + 2$

$f'(x) = 3x^2 + 10x + 2$

α, β

critical point

$f: \mathbb{R} \rightarrow \mathbb{R}$

$f: \mathbb{R} \rightarrow \mathbb{R} \quad S = \{\alpha, \beta\}$

$f'(\alpha) = 0$

$f''(\alpha)$

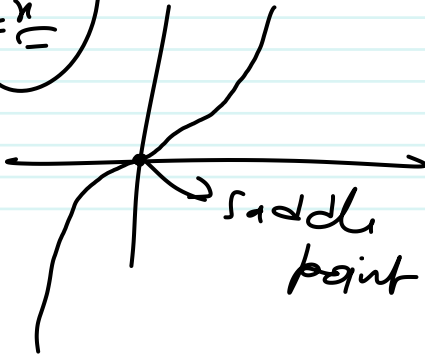
$f''(\alpha) > 0$



$f''(\alpha) < 0$ Local minima
Local maxima

Maxima & minima & saddle point

$f(x) = \frac{x^3}{3}$



$$f''(x) = 0$$

saddle point

$$f''(x) = 0$$

$$f(x) = x^3$$

$$f'(x) = 3x^2 = 0 \Rightarrow x = 0$$

0 is a min

$$f' = 6x$$

$$f'(0) = 0$$

$$f''(x) = 6$$

$$f(x)$$

$$(-1, 1)$$

$$f'(0) = 0$$

$$f'(x) = 5x^4 > 0$$

$$(-1, 1)$$

$$f''(0) = 0$$

$$f'(x) > 0$$

$$(-1, 1)$$

$$f'(x) = 0$$

$$f(x) = \frac{\ln(x)}{x}$$

$$(0, \infty)$$

$$f'(x) = \frac{x \cdot x - \ln x \cdot 1}{x^2}$$

$$= \frac{1 - \ln x}{x^2} = 0$$

$$\Rightarrow \ln x = 1$$

e critical point

$$f(x) = \frac{x^2 - 1}{x - 1}$$

$$x \neq 1$$

$$x = e$$

$$f(x) = \begin{cases} \frac{\ln(x)}{x} & x \neq 0 \\ 2 & x = 0 \end{cases}$$

$$x = 1$$

$$(0, \infty)$$

$$[0, \infty)$$

$$e,$$

$$\lim_{n \rightarrow 0^+} \frac{\ln(n)}{n}$$

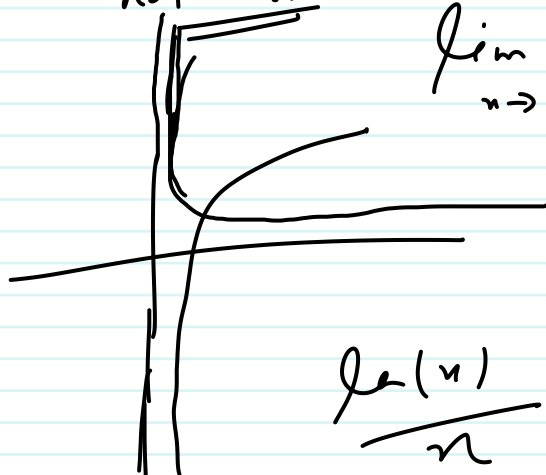
not exist

$$\lim_{n \rightarrow 0} \frac{\ln(1+n)}{n} = 1$$

$$\lim_{n \rightarrow 0^+} \frac{\ln(n)}{n} = \ln(n) \times \frac{1}{n}$$

$$\frac{\sin}{n} = 1$$

$$-\infty \times \infty$$

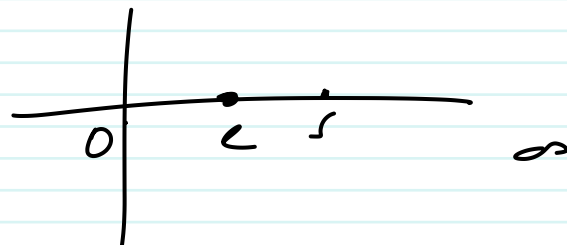


$$f(x) = \frac{\ln x}{x}$$

(0, \infty)

(1, e)

$$\frac{\ln x}{x}$$



$$f'(x)$$

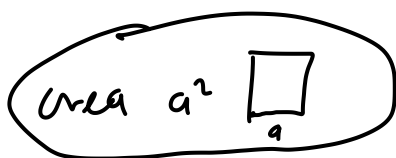
$$f'(x) = \frac{1 - \ln x}{x^2}$$

$$f''(x) = \frac{-\frac{1}{x} \cdot x^2 - (1 - \ln x) \cdot 2x}{(x^2)^2}$$

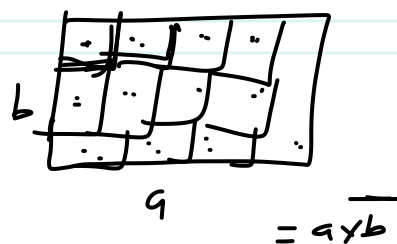
$$f''(x) = -\frac{(x + 2x(1 - \ln x))}{(x^2)^2}$$

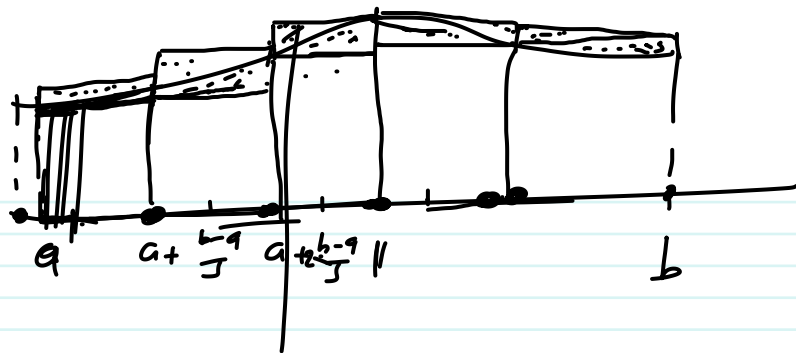
$$f''(e) = \frac{-(e + 2e(0))}{(e^2)^2} = \frac{-e}{e^4} < 0$$

Riemann



4





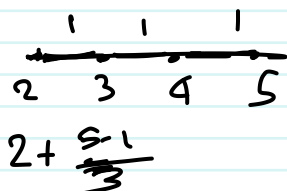
$$\frac{[a, b]}{5} \quad \text{in } 5$$

$$f(n) =$$

$$[2, 5]$$

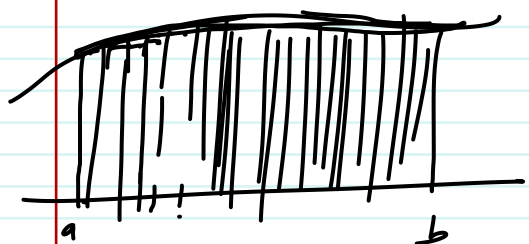
$$a + \frac{b-a}{5}$$

$$\left(\frac{b-a}{5} \right)$$



$$\left(\frac{5-2}{3} \right) = \frac{3}{3} = 1$$

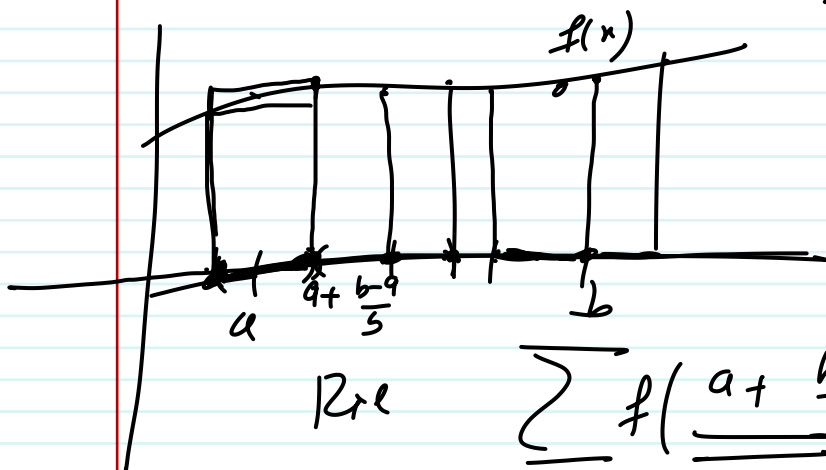
a



$$\frac{b-a}{1000}$$

$$\lim_{n \rightarrow \infty} \frac{b-a}{n}$$

$$n \rightarrow \infty$$



$$\text{Riemann } \sum f\left(\underline{a + \frac{b-a}{5}}\right) \times \left(\frac{b-a}{5}\right) \neq$$

$$\text{Let } \sum f(a) \times \frac{b-a}{5} =$$

$$\lim_{n \rightarrow \infty} \sum \text{let } t \times \frac{b-a}{n}$$

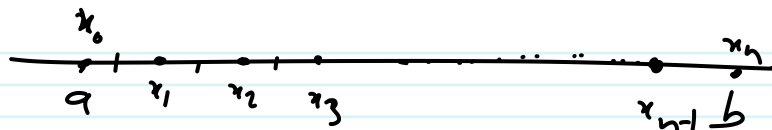
$$= \lim_{n \rightarrow \infty} \text{Riemann } \times \frac{b-a}{n} =$$

$$\text{exactly area} \int_a^b f(x) dx$$

$$\underline{\underline{\int x^2 dx = \frac{x^3}{3}}}$$

$$[a, b]$$

$$\int_a^b x^2 dx = \frac{x^3}{3} \Big|_a^b = \frac{b^3}{3} - \frac{a^3}{3}$$



$$\underline{\underline{ISI}}$$

$$[x_0, x_1], [x_1, x_2], \dots, [x_{n-1}, x_n]$$

$$[x_0, x_1], [x_1, x_2], \dots, [x_{n-1}, x_n] = \|P\| = \frac{b-a}{n}$$

Left

$$\lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} f(x_i) \cdot \frac{b-a}{n}$$

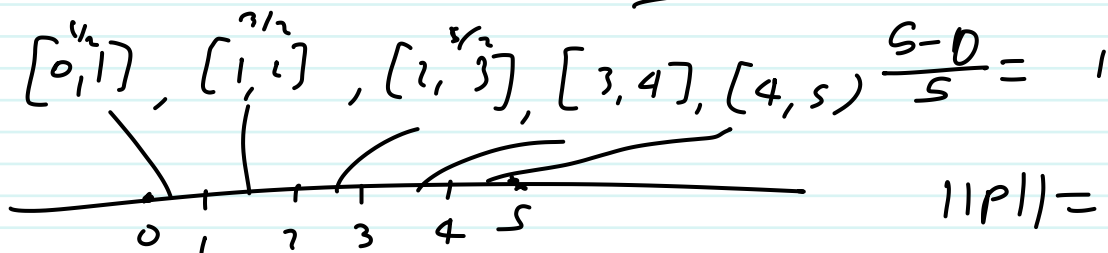
Right

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \cdot \frac{b-a}{n}$$

$$\underline{\underline{f(x) = x^2}}$$

$$\underline{\underline{[0, 5]}}$$

$$\underline{\underline{5 \text{ sub}}}$$



$$\|P\| = \frac{5-0}{5} = 1$$

$$\text{Left Rie Sum} = \underline{f(0)} \times 1 + \underline{f(1)} \times 1 + \underline{f(2)} \times 1 + \underline{f(3)} \times 1 + \underline{f(4)} \times 1$$

$$= 0 \cdot 1 + 1 \cdot 1 + 4 \cdot 1 + 9 \cdot 1 + 16 \cdot 1$$

$$= 1 + 4 + 9 + 16$$

$$=$$

